Lucas A. Saavedra

Contact and Personal Information

Full name Lucas Ariel Saavedra
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Education

2018 – 2023 **Computer Engineering**, Pontifical Catholic University Argentina (UCA), Buenos Aires, Argentina.
Final work: *Comparison of Machine Learning models for the analysis of anomalous diffusion of acetylcholine neurotransmitter receptor trajectories.*
GPA: *8.02/10.*

Research experience

January 2022 – To date **Research Assistant.** Molecular Neurobiology Laboratory, BIOMED UCA-CONICET, (Prof. Francisco J. Barrantes, head), Pontifical Catholic University Argentina (UCA), Buenos Aires, Argentina.

- Implementation of Deep Learning models for the analysis of static and dynamic properties of the receptor for the neurotransmitter acetylcholine (nAChR) with orientation to graph theory, artificial neural networks and geometric computing.
- Analysis of distribution and density of nicotinic acetylcholine receptors (nAChRs) in the plasma membrane of CHO-K1/A5 cells using super-resolution microscopy (STED and STORM). Advanced analytical methods implemented with MATLAB, based on graph theory and computational geometry to study the topography of nanoclusters.
- Development of efficient and autonomous algorithm based on binary classifications with supervised Graph Neural Networks (GNNs) to detect and quantify the dynamic clustering of particles in membrane proteins in single-molecule microscopy (SMLM) data. The algorithm was developed with TensorFlow.
- Deep Learning-based analysis of single-particle tracking datasets obtained with STORM and MINFLUX.
- Application of AlphaFold3-equivalent models to study protein-ligand structures (specifically, the nAChR and cholesterol), combined with molecular dynamics (MD) simulations using the GROMACS software and high-performance computing (HPC) platforms with SLURM.

Research experience (continued)

January 2025 – June 2025

Research Assistant. Department of Data Science, Data Science and AI Laboratory, Pontifical Catholic University Argentina (UCA), Buenos Aires, Argentina.

- Supervisor: Prof. Antonio Quintero-Rincón
- Analysis of electroencephalogram (EEG) data during mental arithmetic tasks using self-supervised learning (specifically, hierarchical variational autoencoders) to detect specific cognitive activity with PyTorch.

Research Publications

Full-Length Publications in International Peer Reviewed Journals

- G. Muñoz-Gil, H. Bachimanchi, J. Pineda, *et al.*, "Quantitative evaluation of methods to analyze motion changes in single-particle experiments," *Nature Communications*, vol. 16, no. 1, p. 6749, 2025, ISSN: 2041-1723 (Electronic) 2041-1723 (Linking). DOI: `10.1038/s41467-025-61949-x`.
- F. Reina, **L. A. Saavedra**, C. Eggeling, and F. J. Barrantes, "Concurrent diffusion of nicotinic acetylcholine receptors and fluorescent cholesterol disclosed by two-colour sub-millisecond minflux-based single-molecule tracking," *Nature Communications*, vol. 16, no. 1, p. 6336, 2025. DOI: `10.1038/s41467-025-61489-4`.
- **L. A. Saavedra** and F. J. Barrantes, "Temporal convolutional networks work as general feature extractors for single-particle diffusion analysis," *Journal of Physics: Photonics*, vol. 7, no. 2, p. 025 017, 2025, ISSN: 2515-7647. DOI: `10.1088/2515-7647/adbec8`.
- **L. A. Saavedra**, A. Mosqueira, and F. J. Barrantes, "A supervised graph-based deep learning algorithm to detect and quantify clustered particles," *Nanoscale*, vol. 16, no. 32, pp. 15 308–15 318, 2024, ISSN: 2040-3364. DOI: `10.1039/D4NR01944J`.
- **L. A. Saavedra**, H. Buena-Maizón, and F. J. Barrantes, "Mapping the nicotinic acetylcholine receptor nanocluster topography at the cell membrane with sted and storm nanoscopies," *International Journal of Molecular Sciences*, vol. 23, no. 18, p. 22, 2022. DOI: `10.3390/ijms231810435`.

Under review

- **L. A. Saavedra** and F. J. Barrantes, "Machine learning in membrane protein single-molecule tracking via superresolution optical microscopy," *Cells*, 2026.

Presentations at International Conferences

- M. F. Colavitta, P. G. Sanz, **L. A. Saavedra**, L. Grasso, and F. J. Barrantes, "Environmental enrichment delays the onset of cognitive decline in a transgenic mouse model of alzheimer disease," in *XL Annual Meeting SAN*, 2025.
- **L. A. Saavedra** and F. J. Barrantes, "Machine learning empowers static and dynamics clustering characterization of transmembrane receptors," in *Building Bridges in Computational Biophysics (BBCB) v3.0*, 2024.
- F. J. Barrantes, A. Mosqueira, H. Buena-Maizón, and **L. A. Saavedra**, "The individual molecule and its tribe, the nanocluster: The case of the nicotinic acetylcholine receptor," in *11th International Weber Symposium*, Punta del Este, Uruguay, 2023.

Teaching experience

March 2024 – June 2024 — **Assistant professor.** Discrete mathematics, Pontifical Catholic University Argentina (UCA), Buenos Aires, Argentina. Main topics: Number theory, Graph theory, Demonstration methods, Boolean algebra.

July 2023 – December 2024 — **Assistant professor.** Data oriented programming, Pontifical Catholic University Argentina (UCA), Buenos Aires, Argentina. Main topics: Object Oriented Programming (OOP), Numpy, Pandas, Python, Java, SQL, Software design with UML.

March 2023 – June 2025 — **Assistant professor.** Introduction to Data Science, Pontifical Catholic University Argentina (UCA), Buenos Aires, Argentina. Main topics: Definitions and concepts of Artificial Intelligence, Introduction to artificial neural networks, Scientific method.

Work experience

May 2025 – To date — **Senior Full-Stack AI Developer.** Accenture, Buenos Aires, Argentina.

- Inside the GenAI Studio team, I develop Large Language Models (LLMs)-based agents for code migration and scientific research. I employ LangGraph, FastAPI, and ElasticSearch for the development of these agents. Also, I participated in computer vision projects.

December 2023 – May 2025 — **Freelancer.** UpWork (Click here to see my profile).

- Web scraping scripts with Python and Selenium. Scripts were hosted in Linux EC2 instances of AWS.
- YOLO training to identify basketball players in video matches from an european basketball league. A SQL database hosted in Supabase was maintained to save information from a handcrafted scraper of the FIBA website.
- Graphical User Interfaces (GUI) developed with built-in Napari plugins to gather information from laser spots pictures provided by the client. This plugins were combined with decision tree classifiers implemented in scikit-learn to extract information from images.

June 2021 – June 2022 — **Back-End Developer.** Deloitte, Buenos Aires, Argentina.

- Participation in Agile methodologies like Scrum.
- Threat Intelligence Back-End Developer.
- I worked developing Codex Gigas, a malware “DNA” profiling search engine discovers malware patterns and characteristics assisting individuals who are attracted in malware hunting, and ImageSimilarity, a tool that divides a group of images according to their similarity grade using the SSIM method. Also, I developed ThreatConnect apps.
- Python, Docker, MongoDB, and SQL were used for development.

January 2021 – June 2021 — **Cyber Risk Advisory Intern.** Deloitte, Buenos Aires, Argentina.

- Offensive/Defensive Security and Threat Intelligence basic concepts

Certificates and Courses

Certifications

2025 **IELTS Score: 7.5**

Duolingo English Test (DET) Score: 135 (Advanced: CEFR C1)

2022 **AWS Certified Cloud Practitioner**

Courses

2025 **Basic to Advanced: Retrieval-Augmented Generation (RAG)**, Udemy

Bioinformatics; Learn Docking & Mol Dynamics Simulation, Udemy

Introduction to Statistics, Stanford Online, Coursera

2024 **Deep Neural Networks with PyTorch**, IBM, Coursera

2022 **Natural Language Processing in TensorFlow**, DeepLearning.AI, Coursera

DeepLearning.AI Tensorflow Developer, DeepLearning.AI, Coursera

Data Science, Coderhouse

2021 **Machine Learning**, Stanford Online, Coursera

Programming Foundations: Design Patterns (2013), Linkedin Learning

R Programming, Johns Hopkins University, Coursera

Docker for the Absolute Beginner - Hands On - DevOps, Udemy

Programming Foundations: Design Patterns (2013), LinkedIn Learning

2020 **The Data Scientist's Toolbox**, Johns Hopkins University, Coursera

Neural Networks and Deep Learning, DeepLearning.AI, Coursera

CS50's Introduction to Artificial Intelligence with Python, Harvard, edX